UTM
UNIVERSITI TEKNOLOGI MALAYSIA

COURSE INFORMATION

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School/Faculty:	**Computing**		
Program name:	**Bachelor in Software Engineering with Honours**		
Course code:	**SCSM2113**	**Academic Session/Semester:**	**2025/2026 /Semester 1**
Course name:	**Human-Computer Interaction Fundamentals**	**Pre/co requisite (course name and code, if applicable):**	**SCSE1203 Software Engineering Principles**
Credit hours:	**3**		

Course synopsis	**This foundational course introduces the principles and practices of Human-Computer Interaction (HCI), focusing on user-centered design and interface development. Students will explore interaction design concepts, usability principles, and evaluation methods. Topics include user requirements gathering, prototyping, usability testing, and cultural considerations in interface design, with exposure to real-world industry practices.The course uses interactive lectures, hands-on prototyping, collaborative projects, and sessions with industry partners.By the end of the course, students will be able to analyze user needs, design and prototype interfaces, and evaluate usability while demonstrating teamwork and cultural awareness in HCI practices.**
Course coordinator (if applicable)	**Assoc. Prof. Ts. Dr. Masitah Ghazali**

Course lecturer(s)	**Name**	**Office**	**Contact no.**	**E-mail**
	Assoc. Prof. Ts. Dr. Masitah Ghazali			**masitah**
	Prof. Ts. Dr. Nor Azman Ismail			**azman**
	Dr. Layla Rasheed Abdallah Hasan			**layla.hasan**
	Dr. Yusnaidi Md Yusof			**yusnaidi.kl**

Mapping of the Course Learning Outcomes (CLO) to the Programme Learning Outcomes (PLO), Teaching & Learning (T&L) methods and Assessment methods:

No.	CLO*	PLO **(MQF Cluster	***Taxonomies and ****Graduate attributes	T&L methods	*****Assessment methods
CLO1	Apply the basic principles of human–computer interaction and different types of user interface	PLO1 (C1)	C3 SC1	Lecture, active learning	PR 1, T, F
CLO2	Design solutions for the assigned problem according to HCI methods and techniques to solve the identified design issues	PLO2 (C2)	C6 SC2	Lecture, problem-based learning	PR 2, PR 3, T, F
CLO3	Participate and contribute actively in team settings (team members and stakeholders) in diverse learning and working communities in developing the community or industry based applications	PLO4 (C3B)	A2 A(SI1)	Problem-based learning, work-based learning (WBL)	PR 4

This is the basic mapping required for the CI. Any added information is allowed (extra columns for weight or other elements) provided this is made consistent for all CI at program/school/faculty level.
*Up to 5 CLO
Refer ***Taxonomies of Learning and ****UTM's Graduate Attributes for measurement of outcomes achievement
*****T – Test; Q – Quiz; HW – Homework; Asg – Assignment; PR – Project; Pr – Presentation; F – Final Exam etc.

****MQF Cluster Code**
C1 = Knowledge & Understanding, **C2** = Cognitive Skills, **C3A** = Practical Skills, **C3B** = Interpersonal Skills, **C3C**= Communication Skills, **C3D** = Digital Skills, **C3E** = Numeracy Skills, **C3F** = Leadership, Autonomy & Responsibility, **C4A** = Personal Skills, **C4B** = Enterpreneurial Skills, **C5** = Ethics & Professionalism

Details on Innovative T&L practices:

No.	Type	Implementation
1	Active learning	Conducted through in-class activities on informal cooperative learning
2	Problem-based learning	Conducted through group project based on real-world problems, which tasks are given in sequential steps throughout the semester. Students in a group of 3-4 are asked to deliver a usable and tested interactive system interface
3	Work-based learning	Industry partner will involve in the workshop and/or critic session

Transferable skills (generic skills learning in course of study which can be useful and utilised in other settings) :
PL4 (C3B) - Able to contribute actively in team settings in diverse learning and working communities

Student learning time (SLT) / Effective Learning Time (ELT) details:

Week/ Meeting	Course Content Outline and Subtopics	CLO* Compulsory	Learning and Teaching Activities									TOTAL SLT
			Face-to-Face (F2F)								Non F2F Independent Learning	
			Physical				Online (Synchronous)				Online (Asynchronous)	Others
			L	T	P	O	L	T	P	O		
Week 1	Introductory to Course Introduction to Human-Computer Interaction ● Good and poor design ● Interaction design vis-à-vis human-computer interaction ● User experience and usability ● The process of interaction design ● Interaction design and the user experience ● HCI major / general design principles &	CLO1	2		1						1		4
Week 2	Users: cognition and emotion ● Cognition: attention, memory, recognition ● Cognitive frameworks: mental models ● Expressive interfaces and positive emotions ● Frustrating interfaces and negative emotions ● Persuasive technologies and behavioural change	CLO1	2		1						2		5
Week 3	Trends in Interface ● Interaction types ● Interfaces types ● Social mechanisms in communication and collaboration	CLO1	2		1						2	2	7
Week 4	The Process of Interaction Design ● Involving users in development ● Principles of user-centred approach ● Four basic activities in interaction design ● Practical issues	CLO1	2		1						2		5

Week 5	Data Gathering (Industrial partner) ● Different kinds of requirements: persona, goals, environments ● Data gathering for requirements types ● Overview of how to plan interviews, design questionnaires and run observations	CLO1	2		1							2	5
Week 6	Data Analysis & Establishing Requirements for Design ● The differences between qualitative and quantitative data ● Analyse the results from data gathering sessions o Develop scenarios o Perform HTA	CLO2 CLO3	2	1								2	5
Week 7	Design ● Metaphors and analogies ● Information architecture ● Navigation ● Visual design principles (Gestalt Principles, Pragnanz Law) ● UI elements	CLO2	2		1							2	5
Week 8	Prototyping ● Prototyping o Wireframe o Low-fidelity o High-fidelity ● Conceptual design	CLO2	2		1							2	5
Week 9	Prototyping (cont.) ● Concrete design o Scenarios in design o Prototypes in design (storyboard, card-based) o Customer journey KIV ● Construction: physical computing, SDK	CLO2 CLO3			3							2	5
Week 10	Evaluation ● Why, what, where and when of evaluation ● Evaluation framework ● Evaluation approach / types: controlled, natural, not involved users ● How to: usability testing, experiment, field studies	CLO2	2		1							2	5
Week 11	Evaluation (cont.) ● Inspection (heuristic evaluation, walkthrough) ● Analytics ● Predictive model (Fitts' Law)	CLO2 CLO3	2		1						2		5
Week 12	Data Analysis, Interpretation and Presentation ● Simple quantitative analysis ● Simple qualitative analysis ● Tools to support data analysis ● Presenting the findings	CLO2	2		1						2		5
Week 13	Re-design ● Translate findings to design improvement	CLO2	2		1						2	2	7
Week 14	HCI Showcase (Industry involvement)	CLO3	3								3	2	8
Week 15		CLO?											0
											SUB-TOTAL SLT :		**76**

	Continous Assessment	%	Face-to-Face (F2F)		NF2F Independent Learning for Assessment		TOTAL SLT
			Physical	Online (Synchronous)	Online (Asynchronous)	Others	
1	Project 1: Proposal report	5			3		3
2	Project 2: Establishing requirements report	10			5		5
3	Project 3: Design and prototype report	20			10		10
4	Project 4: Evaluation report	20	4			6	10
5	Test	15	1			3	4
					SUB-TOTAL SLT :		32

	Summative Assessment	%	Face-to-Face (F2F)		NF2F Independent Learning for Assessment		TOTAL SLT
			Physical	Online (Synchronous)	Online (Asynchronous)	Others	
1	Final Exam	30	3			9	12
2							0
3							0
4							0
5							0
					SUB-TOTAL SLT :		12
					SLT for Assessment:		44
					GRAND TOTAL SLT:		120

A	% SLT for F2F Physical Component	41.67
B	% SLT for Online & Independent Learning Component :	58.33
C	%SLT for Online Component:	28.33
D	% SLT for All Practical Component:	11.67
D1	% SLT for F2F Physical Practical Component:	11.67
D2	% SLT for F2F Online Practical Component:	0.00
Please tick (/) if this course is Industrial Training/ Clinical Placement/ Practicum using 50% of Effective Learning Time (ELT)		

Identify special requirement or resources to deliver the course (e.g.,software,nursery, computer lab, simulation room etc)

References (include required and further readings, and should be the most current)

Rogers, Y., Sharp, H., and Preece, J., (2023) "Interaction Design: Beyond Human-Computer Interaction", 6th. Edition, New York: J. Wiley & Sons.

Additional references

1. Christoper, R. B. (2020), Learn Human-Computer Interaction: Solve human problems and focus on rapid prototyping and validating solutions through user testing, Packt Publishing.
2. Samuel Wallace (2020, Handbook of Human-Computer Interaction, 1st. Edition, Walford Press.

Online
http://elearning.utm.my

Other additional information (if applicable)
Academic honesty and plagiarism: *(Below is just a sample)*
Assignments are individual tasks and NOT group activities (UNLESS EXPLICITLY INDICATED AS GROUP ACTIVITIES) Copying of work (texts, simulation results etc.) from other students/groups or from other sources is not allowed. Brief quotations are allowed and then only if indicated as such. Existing texts should be reformulated with your own words used to explain what you have read. It is not acceptable to retype existing texts and just acknowledge the source as a reference. Be warned: students who submit copied work will obtain a mark of zero for the assignment and disciplinary steps may be taken by the Faculty. It is also unacceptable to do somebody else's work, to lend your work to them or to make your work available to them to copy.

Other additional information (if applicable)

1. To pass this course, students must pass both Continuous and Final assessment.

2. All students must adhere to the following to pass a computing course:
i. Pass both continuous assessment AND final assessment
ii. Get at least minimum pass marks following UTM regulation (40%)
- The passing mark for Continuous Assessment is 30% of continuous marks.
- The passing mark for the Final Assessment is 20% of the final marks.-
- If a student does not achieve the passing mark for continuous assessment or the passing mark for the final assessment, then the student's overall final mark will be set below the UTM passing mark.
Attendance is compulsory and will be taken in every lecture session. Student with less than 80% of total attendance is not allowed to sit for final exam.

3. Students are required to behave and follow the University's dressing regulation and etiquette all the time.

4. Exercises and tutorial will be given in class and some may be taken for assessment. Students who do not do the exercise will lose the coursework marks for the exercise.

5. Assignments must be submitted on the due dates. Some points will be deducted for late submissions. Assignments submitted three days after the due date will not be accepted.

6. Make up exam will not be given, except to students who are sick and submit medical certificate confirmed by UTM panel doctors. Make up exam can only be given within one week of the initial date of exam.

Disclaimer:

All teaching and learning materials associated with this course are for personal use only. The materials are intended for educational purposes only. Reproduction of the materials in any form for any purposes other than what it is intended for is prohibited.
While every effort has been made to ensure the accuracy of the information supplied herein, Universiti Teknologi Malaysia cannot be held responsible for any errors or omissions.

ELT = (Theory + Industrial Guidance + Assessment) x 50%

Total of credit for LI/Practical = ELT/40 Notional Hours
Note: For ODL Programme : Courses with mandatory practical requirement imposed by programme standards or any related standards can be exempted from complying to the minimum 80% ODL delivery rule in the SLT.

Prepared by:	**Certified by:**
Name: Assoc. Prof. Ts. Dr. Masitah Ghazali	**Name:** Radziah Mohamad
Signature:	**Signature:**
Date: 2 Sept. 2025	**Date:** 5/10/2025